

Course: Deep Learning (DL) and its Application in Natural Language Processing (NLP)

Course Code: DA345

Course description:

This course presents an introduction to the **Deep Learning (DL)** and its application in **Natural Language Processing (NLP)**. We will explore theoretical foundation of DL and their practical implementation to solve different NLP related tasks such as *text classification, sentiment analysis, machine translation, text summarization, text generation*, etc..

Prerequisite (s):

Student should have some knowledge in

- Mathematics: *Linear Algebra, Multivariate Calculus, Basis Optimisation and Basic probability*
- Computer programming: *Any one from C/C++/Python (recommended for the class project and assignments)/MATLAB/Octave*
- Basic concept in Algorithms and Data Structure
- Introduction to Machine Learning

Credit Hours: 4, approximately 60 credit hours

Text(s):

- Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning.*, MIT Press, 1st edition, 2016. online
- Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola. *Dive into Deep Learning*, Cambridge University Press, 2023. online
- Simon J.D. Prince. *Understanding Deep Learning*, MIT Press, 2023. online
- Eugene Charniak. *Introduction to deep learning*, MIT Press, 2018.
- Michael Nielsen. *Neural Networks and Deep Learning*, online
- Dan Jurafsky and James H. Martin. *Speech and Language Processing.* draft, 3rd edition, 2024. [online]
- Delip Rao, Brian McMahan. *Natural Language Processing with PyTorch*, O'Reilly Media, Inc, 2019
- Lewis Tunstall, Leandro von Werra, and Thomas Wolf. *Natural Language Processing with Transformers Book*, O'Reilly Media, Inc, 2022. online code only
- Yoav Goldberg. *A Primer on Neural Network Models for Natural Language Processing.* online

Course Objective:

Students will get to know:

- the core concepts of DL, including network design, training and evaluation.
- common DL architectures for tasks like object recognition, image generation, sentiment analysis, text summarization, and machine translation, etc..
- techniques for text preprocessing to clean and prepare textual data for NLP tasks.
- advanced deep learning models for different text processing tasks in NLP
- use of different DL and NLP toolkits (like PyTorch, NLTK, PyTorch) to solve practical NLP tasks.
- the performance of DL models and identify potential biases or errors.

Course outcomes:

After successful completion of this course, students will be able to:

- understand the mathematical basis of various deep-learning models.
- explain key deep learning concepts such as activation functions, loss functions, backpropagation, regularization, etc..
- design, implement, and evaluate feedforward, convolutional, recurrent neural networks and modern transformer based architectures.
- explain textual data and extract representative features.
- solve practical problems in NLP using deep learning methods.
- gain hands-on experience with deep learning models in PyTorch ¹.

Grade Distribution:

Quizzes and assignments	25%
Midterm Exam	10%
Class project	25%
Final Exam	40%

Course outline (tentative) and syllabus:

The tentative syllabus is as follows:

- Artificial neural network (ANN): Modelling Single neuron activity, different types of activity functions (sigmoid, tanh, ReLU, ELU etc.), how to connect multiple neurons to form a network, Multi-layer perceptron
- Optimization: Back propagation, different loss functions, gradient decent, stochastic gradient decent and different update rules (AdaGrad, RPMSprops, Adam etc.) for network parameters, regularization, dropout, batch normalisation etc.
- Deep learning toolbox: Explore a deep learning toolbox like PyTorch (my personal choice)/ TensorFlow and their autograd functionalities

¹<https://pytorch.org/>

- Convolutional neural network (CNN): Concept of kernel and convolution, some pooling operation (max, average etc.), some standard CNN architectures like LeNet, AlexNet, VggNet, ResNet etc. and concept of transfer learning
- Recurrent neural network (RNN): Sequential data and how to handle those using neural network, general RNN architecture, some popular RNN architectures like *Long short-term memory (LSTM)*, *Gated recurrent unit (GRU)* and their different variants
- Deep generative models: *Variational Autoencoders (VAE)*, *Generative Adversarial Networks (GAN)*, *Normalizing flows*, *Diffusion models*, etc.
- Neural language model:
 - Introduction to NLP
 - Text preprocessing: tokenisation, stop words, stemming, lemmatisation, etc.
 - Vector representations of text: *Bag of Words*, *TF-IDF*, *word embeddings*, *Word2Vec*, *GloVE*, etc.
 - Sequence modelling: *Recurrent neural network (RNN)*, *Self-Attention network*, etc.
 - Transformers: *Attention*, *BERT* and its different variants, Encoder-Decoder models
 - Large language model (LLM): *GPT* different variants, *pre-trained language model*, *transfer learning*
 - Application: *text classification*, *sentiment analysis*, *Named Entity Recognition (NER)*, *machine translation*, *text summarization*, *text generation*, etc.

The weekly coverage might change as it depends on the progress of the class. However, you must keep up with the reading assignments. Each week assumes 4 hour lectures. Reading assignments are from reference textbook will be mentioned in the suggested reading material in the *course url*.

Week	Content
Week 1	• Motivation: Artificial neural network (ANN): Modelling Single neuron activity, different types of activity functions (sigmoid, tanh, ReLU, ELU etc.), how to connect multiple neurons to form a network, Multi-layer perceptron
Week 2	• Optimization: Back propagation, different loss functions, gradient decent, stochastic gradient decent and different update rules (AdaGrad, RPMSprops, Adam etc.) for network parameters, regularization, dropout, batch normalisation etc. • Assignment-1
Week 3	• Deep learning toolbox: Explore a deep learning toolbox like PyTorch (my personal choice)/ TensorFlow and their autograd functionalities • Assignment-2
Week 4	• Convolutional neural network (CNN): Concept of kernel and convolution, some pooling operation (max, average etc.), some standard CNN architectures like LeNet, AlexNet, VggNet, ResNet etc. and concept of transfer learning • Assignment-3 • Class project discussion
Week 5	• Recurrent neural network (RNN): Sequential data and how to handle those using neural network, general RNN architecture, some popular RNN architectures like <i>Long short-term memory (LSTM)</i>, <i>Gated recurrent unit (GRU)</i> and their different variants • Assignment-4
Week 6	• Class project proposal • Deep generative models: <i>Variational Autoencoders (VAE)</i> • Assignment-5
Week 7	• Deep generative models: <i>Generative Adversarial Networks (GAN)</i>, <i>Normalizing flows</i>, <i>Diffusion models</i>, etc. • Course review for the mid-term examination
Week 8 - 15	• Neural language model: – Introduction to NLP – Text preprocessing: tokenisation, stop words, stemming, lemmatisation, etc. – Vector representations of text: <i>Bag of Words</i>, <i>TF-IDF</i>, <i>word embeddings</i>, <i>Word2Vec</i>, <i>GloVE</i>, etc. – Sequence modelling: <i>Recurrent neural network (RNN)</i>, <i>Self-Attention network</i>, etc. – Transformers: <i>Attention</i>, <i>BERT</i> and its different variants, Encoder-Decoder models – Large language model (LLM): <i>GPT</i> different variants, <i>pre-trained language model</i>, <i>transfer learning</i> – Application: <i>text classification</i>, <i>sentiment analysis</i>, <i>Named Entity Recognition (NER)</i>, <i>machine translation</i>, <i>text summarization</i>, <i>text generation</i>, etc. • Assignment-6 & 7 • Course review for the final examination